

# VRDD: Vehicular Relevance based Data Dissemination in Vehicular Ad-hoc Network

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**Abstract:** Safety message dissemination in VANET needs to be fast as it allows vehicles to be alert and avoid adverse situations. A safety message gets disseminated within a topology by multi-hop broadcast techniques and existing protocols apply different controlled rebroadcast mechanisms to address the issues of multi-hop broadcast namely “packet loss” and “dissemination delay”. But no work has been done to take care the direction of message dissemination. In the real scenario, all the vehicles within an topology of interest are not equally affected by an emergency incident. The road where an incident happens gets most affected along with the roads connected by nearby junctions. Vehicles along these roads are more relevant w.r.t this incident, than the other vehicles and should receive the safety message as fast as possible with higher priority. Hence, this paper proposes a novel method to predict the relevance of vehicles using a distributed algorithm which heuristically tries to disseminate safety message towards the relevant vehicles. We have compared our work with three state of art algorithms and found that VRDD is compatible with the existing algorithms on the metrics of packet loss and dissemination delay but clearly leading on the metric of relevant dissemination.

## 1 Introduction

Vehicular Ad-hoc Network (VANET) has become a wide area of research to provide vehicles safety and infotainment services recognized by Intelligent Transportation Systems (ITS) [1]. Vehicles are pre-installed with a radio device named “On Board Unit (OBU)” by which they communicate among themselves and also communicate with static roadside infrastructure named “Road Side Unit (RSU)”. VANET can form both infrastructure network as well as ad-hoc network. Hence, there are two types of communication possible in VANET namely “Vehicle to Infrastructure (V2I)” and “Vehicle to Vehicle (V2V)”. Though some protocols [2] [3] relies on V2I communication for data dissemination, RSU deployment is costly and may be unavailable in many places. So we rely on infrastructure less ad-hoc V2V communication only for dissemination of safety messages. V2V Communications within vehicles can be of three types viz. One to Many (e.g. advertising the availability of a service by a vehicle), Many to Many (e.g. disseminating a safety message among vehicles), and One to One (e.g. availing Internet service or other infotainment services). In Many to Many communication, multi-hop broadcast technique is used to disseminate an important message to as many vehicles as possible within an Area of Interest (AOI). Multi-hop broadcast usually suffers from some well known problems like Broadcast Storm and Network Partition. In existing works, one or both of the challenges have been addressed by applying different techniques. Broadcast Storm problem is usually handled by controlled flooding, where few vehicles are chosen for rebroadcasting to propagate the message. Network Partition problem is handled by choosing vehicle(s) which is predicted to have minimum delay to meet a vehicle from another network fragment.

### 1.1 Motivation

Existing algorithms for handling broadcast storm problem rely on parameters like the location of the vehicle, distance from the sender, channel congestion, number of neighbors, etc. to choose next vehicle for rebroadcasting a Safety Message (SM). But, in an urban scenario, a SM can get propagated towards any direction within AOI

which may not be an important or relevant direction with respect to the incident location. Vehicles traveling along the roads and heading towards incident location are more relevant to the SM than the vehicles which are traveling on different roads or moving away from the incident location. So, if an irrelevant vehicle is chosen as next hop to rebroadcast, it further propagates the SM towards irrelevant direction whereas some important directions may get ignored which will cause some relevant vehicles to remain uninformed about the incident. This is illustrated in Fig. 1. After V1 and V2 met with an accident, observer vehicle V5 originates a Safety Message (SM) and broadcasts it. As V3 and V5 are most affected by the incident, SM needs to propagate along the travel path of V3 and V5 with highest priority so that other vehicles along these paths can be aware of the incident. Vehicle V7 is partially relevant because it is currently not on the affected road, but it is likely to meet the affected location soon. V4 will remain unaffected with the incident because, as based on its current travel direction, it may be predicted that the road it is traveling on will meet the affected road in a far distant location from the incident location. V8 is also unaffected because it is moving away from the incident location. As V8 is one of the most distant vehicles in respect to the communication radius of observer V5, it may get selected as the next rebroadcast node by existing algorithms. Hearing the redundant safety message from V8, relevant vehicle V6 and partially relevant vehicle V7 will cancel their rebroadcast and hence the safety message may not get propagated along the path of V5 and V7 and instead will get propagated along the irrelevant direction of V8 which is clearly unacceptable.

### 1.2 Novelty

The novelty of our proposed protocol named “Vehicular Relevance based Data Dissemination in VANET (VRDD)” is that it aims to disseminate a SM towards the direction of relevant vehicles with higher priority with minimized delay and packet loss. A heuristic method has been used in VRDD to predict the relevance of vehicles and to choose a suitable relevant vehicle for next rebroadcast. As mobility of vehicles are bounded by the road topology, we have assumed that, if a road is not too much irregular, the travel direction of a vehicle on

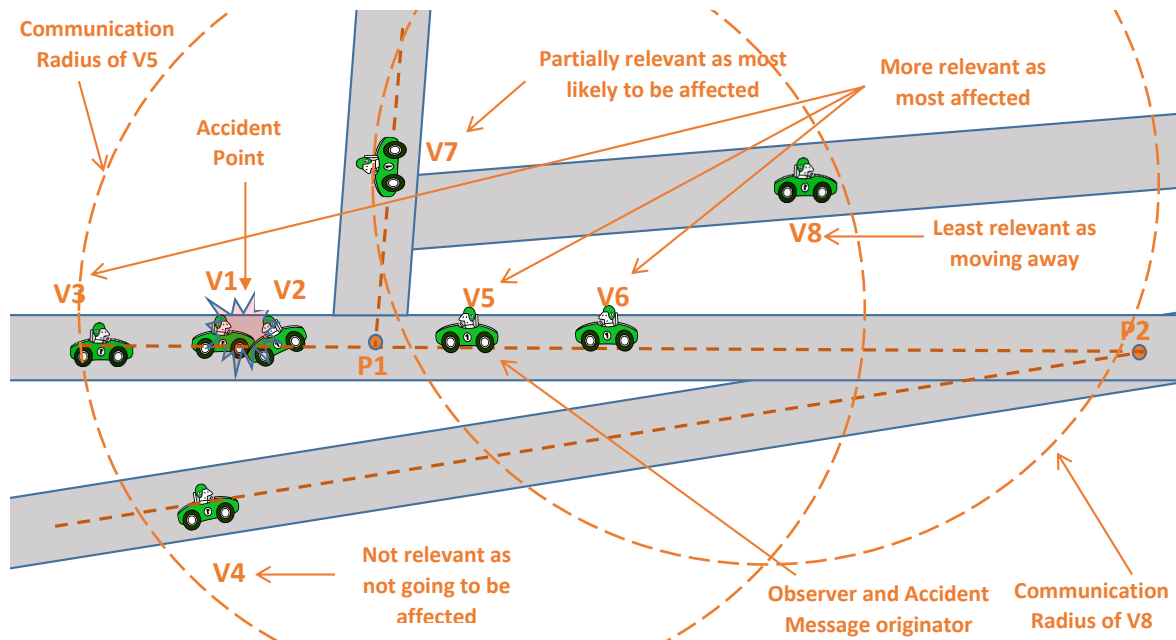


Fig. 1: Illustration of vehicle relevance

that road will match with the travel direction of other nearby vehicles on that same road and will differ with the travel direction of vehicles traveling on different roads. Depending on the deviation of travel direction between sender and receiver vehicle, the algorithm predicts how likely it is for a vehicle to reach nearby the immediate sender's location. If it is predicted that the receiver vehicle is unlikely to reach immediate sender's nearby location, the receiving vehicle is not relevant to that SM and will not be chosen as next hop rebroadcast node. SM is disseminated along relevant directions with higher priority and Broadcast Storm problem is mitigated because only a few relevant vehicles are selected as next hop rebroadcast node. In VRDD, Network Partition problem is handled along each road where the last vehicle of each road waits until it meets a new vehicle along that road to rebroadcast the SM.

The performance of proposed VRDD is measured in one type of Highway topology and two types (irregular and grid) of Urban topology with the metrics of "Packet Loss", "Packet Redundancy", "Dissemination Delay", "Vehicle Coverage Ratio" and "Relevant Vehicle Penetration Ratio". The effectiveness of the proposed algorithm is evaluated and compared with some benchmark state of art algorithms like "TrAD" [14], "DRIVE" [17] and "DV-CAST" [15]. The results show that the proposed VRDD algorithm effectively solves the Broadcast storm and Network Partitioning problem of many to many communication in VANET and delivers SM to most of the relevant vehicles with lesser delay, and packet loss.

Rest of this paper is organized as follows: In Section 2, we have briefly discussed about the IEEE standard "WAVE" for VANET communication and some challenges in VANET data dissemination. Section 3, focuses on some benchmark works done to mitigate the challenges of multi-hop broadcasting in VANET. In Section 4, we have described our proposed VRDD algorithm and its methodologies in details. Section 5 contains comparative performance analysis of simulation results with some benchmark algorithms. Finally in Section 6, the conclusions and future extendibility of VRDD is discussed.

## 2 Background

This section briefly highlights the IEEE 1609 "WAVE" standard and key challenges of data dissemination, in the following two subsections:

### 2.1 IEEE Standard for VANET

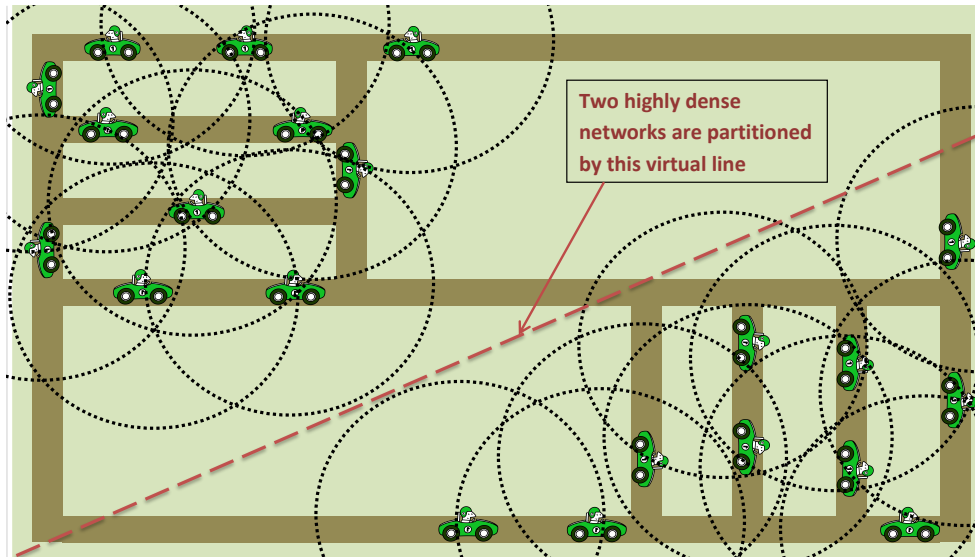
Radio communication among vehicles in VANET uses the standard Wireless Access in Vehicular Environments (WAVE) defined in IEEE 1609 [4]. WAVE recommends the use of seven radio channels of 10 MHz bandwidth (ranges from 5.855 GHz to 5.925 GHz) for "WAVE Short Message (WSM)" packet transmission. The channels are numbered from 172 to 184 where channel 178 (5.89 GHz) is the Control Channel (CCH) used for Timing Advertisement, WAVE Service Advertisement (WSA), etc and channel 172 (5.86 GHz) is used for exchanging Safety Messages (SM). Other channels are used for providing infotainment services as advertised by WSA in CCH. So a vehicle needs to monitor in CCH for any new WSA of its interest even while it is participating in an ongoing service in one of the SCHs. The multi-hop broadcast V2V data dissemination faces few challenges which are described in the next subsection.

### 2.2 Data Dissemination Challenges in VANET

SM disseminating within AOI of VANET using V2V multi-hop broadcast is challenging due to the highly dynamic nature of the network topology. Because of the high velocity, vehicles frequently leave one network and join another by changing roads. In case of traffic congestion, density of vehicles becomes very high and forms a very dense network, whereas in case of light traffic, vehicles are too far apart from each other to form a well connected network. This section describes some major challenges faced in SM dissemination in both highly dense and sparse scenarios.

- **Challenges in Dense Scenario:** In a dense scenario, after receiving a SM, if all vehicles try to rebroadcast it simultaneously, most of the packets will collide with each other and will be lost. This phenomenon is called Broadcast Storm problem (as illustrated in Fig. 2). To mitigate this challenge, various algorithms apply different techniques to de-synchronize the rebroadcast schedule within nearby vehicles.

- **Challenges in Sparse Scenario:** In sparse scenario, vehicles are fragmented into number of disconnected network. Hence, a SM generated for an emergency incident fails to propagate from one fragment to another. This is described as Network Partition problem and is also shown in Fig. 2. Various algorithms choose one or more suitable vehicles and assign the role to carry forward the SM from one network fragment to another.



**Fig. 2:** Illustration of Broadcast Storm and Network Partition Problem

### 3 Related Work

This section describes various algorithms proposed to mitigate the challenges mentioned in the above section in case of V2V ad-hoc network. Controlled flooding or Broadcast Suppression techniques for mitigation of broadcast storm problem in V2V multi-hop data dissemination in VANET has been broadly classified into two classes namely “Sender-oriented Schemes (SOS)” and “Receiver-oriented Schemes (ROS)” [5]. There are certain merits and demerits in both the schemes which are discussed as follows:

In SOS, the SM sender vehicle chooses one or more vehicles among its neighbors and assigns rebroadcast task to them. “Urban Multi-hop Broadcast (UMB)” [6] and “Smart Broadcast (SB)” [7] are examples of SOS where a handshaking mechanism is used to establish a connection between sender and farthest receiver(s) and the sender vehicle explicitly delegates the rebroadcast task to the connected receiver(s). But the connection is not stable as the furthest vehicles can move out of communication range or some obstacle may disrupt the connection before handshaking process can take place. In this case, the sender vehicle again establishes new connection to another vehicle(s), which causes extra delay in SM propagation. Another SOS algorithm “The Traffic Adaptive Data Dissemination (TrAD)” [8] doesn’t establish any connection with receivers. Instead, it implicitly selects vehicles as next hop by calculating priority for all of its neighbors and appending the priority rank list along with the re-broadcast packet of SM. Receiving a SM, all vehicles check their rank in the attached list and assign rebroadcast delay proportional to its rank. SOS algorithms perform well in respect to packet loss, as only the selected vehicles get chance to rebroadcast. But in highly dense scenario, sender vehicle gets overburdened with heavy computation which increases delay.

In ROS, all the receiving vehicles decide for themselves in a distributive manner whether to rebroadcast a SM and if yes, it also decides a wait time or delay before rebroadcast. ROS algorithms can be further categorized in two subclasses called “Probability Based” and “Delay Based” as mentioned in [9]. Vehicles in probability based algorithms [10] [11] [12] rebroadcast a SM with some calculated probability. As not all the vehicles decide to rebroadcast every time, flooding is avoided. “DV-CAST” [13] is an example of probability based algorithm proposed for only for highway scenario uses ‘p-persistence’ method for broadcast suppression. CPB [14] is another probability based algorithm applicable on highway scenario. In CPB, suitable vehicles depending on their position and travel direction form clusters and a cluster head (CH) is elected within each cluster. Each vehicle relays a received message to CH with calculated probability depending on the local vehicular density. In delay based algorithms, vehicles calculate their own rebroadcast delay in

a purely distributive manner based on some parameters like location of the vehicle, distance from the sender, channel congestion, number of neighbors, etc. The vehicle with minimum delay rebroadcasts the SM and if neighboring vehicles of the sender receive this duplicate SM, then they cancel their rebroadcast schedule to avoid flooding. “UV-CAST” [15] algorithm is a delay based algorithm which has been proposed for two dimensional urban scenario. “UV-CAST” gives priority to the vehicles which are near any of the junction within AOI. Thus it requires to know all the junction points in prior which may not be available in few situations. Both in DV-CAST and UV-CAST, vehicles are aware of their local topology by exchanging beacon message and maintain a neighbor list. SEAD [16] is a hybrid algorithm which applies both delay based and probability based technique, but is not applicable for irregular urban scenarios.

DRIVE [17] algorithm works on both highway and urban scenario and doesn’t require vehicles to have knowledge about their local topology. Here, the communication area of the sender vehicle is divided into four quadrants and some area within each quadrant is assumed as “Sweet Spot”. Receiving vehicles calculate whether it belongs to any of the Sweet Spots in respect to sender’s location and if yes, it assigns shorter delay for itself in compared to the vehicles not situated in any of the Sweet Spots. Within the vehicles in Sweet Spots, the vehicles which are far apart from the sender vehicle, get higher priority for rebroadcast. ICARUS algorithm [18] has used the same Sweet Spot concept for controlling flooding and additionally, proposed the method named “Re-routing” to find alternative path in case of traffic congestion. ROS algorithms usually have shorter dissemination delay compared to SOS as the rebroadcast delay computation is distributive, but in highly dense scenario, it suffers from more packet loss compared to SOS algorithms as many vehicles can end up calculating similar rebroadcast delay for themselves.

Network Partition issue has been handled by DV-CAST algorithm [13] for highway scenario only. With the help of periodic beacon message, vehicles categorize their neighbors into three groups namely NB\_FRONT (neighbors moving ahead in same direction), NB\_BACK (neighbors following in same direction) and NB\_OPPOSITE (neighbors moving in opposite direction). Depending on the cardinality of these sets, each vehicle defines its connectivity state with the help of three flags named MDC, ODC and DFlg. Based on the flag combination it determines whether to perform the role of relay node. TrAD algorithm [8] handles Network Partition issue by selecting one or more Store Carry Forward (SCF) agent(s). In TrAD, a vehicle takes up the role of SCF agent if either it is near to any of the junction points or it is moving in the message forwarding direction and is the furthest from the sender. But here the disadvantage is that the algorithm needs to know the list of all junction points of an AOI in prior which may not be feasible



**Table 1** Comparative study among existing algorithms and VRDD

| Algorithm       | Type              | Computation Overhead | Packet Loss | Network Partition | Supported Topology | Relevant-Dissemination |
|-----------------|-------------------|----------------------|-------------|-------------------|--------------------|------------------------|
| Smart Broadcast | Centralized (SOS) | Moderate             | Low         | Not supported     | Urban              | No                     |
| TrAD            | Centralized (SOS) | High                 | Low         | Supported         | Urban/Highway      | No                     |
| DV-CAST         | Distributed (ROS) | Moderate             | Moderate    | Supported         | Highway            | No                     |
| DRIVE           | Distributed (ROS) | Low                  | High        | Supported         | Urban/Highway      | No                     |
| VRDD            | Distributed (ROS) | Moderate             | Low         | Supported         | Urban/Highway      | YES                    |

in all cases. UV-CAST algorithm [15] detects the border vehicles by its proposed “Gift-wrapping” algorithm and selects them as SCF agents. But none of the algorithms ensures that the vehicles selected as SCF agent or Network Bridge vehicle will meet a relevant vehicle to rebroadcast the SM so that it gets further disseminated among other relevant vehicles of another network fragment.

### 3.1 Drawbacks in existing methods and proposed solution

Existing algorithms measure the candidature of a vehicle for becoming a next hop node by primarily focusing on location (distance from sender, distance from any junction point, sweet spot, etc.) or by the help of randomness. But none of them has tried to measure the candidature by relevance of vehicles and hasn’t targeted the dissemination of SM towards the directions of relevant vehicles. If an irrelevant vehicle gets higher priority for rebroadcast than a relevant vehicle, a relevant vehicle may cancel its rebroadcast schedule hearing duplicate SM rebroadcast from irrelevant vehicle. Hence, SM will not be propagated along the path of relevant vehicles and thus, SM may fail to serve its purpose. The situation is described in Section 1.1.

To solve this issue, in [19], we have proposed the “TAFDRV” (first of its kind) algorithm where a simple approach is used to predict the relevance of a vehicle by the use of its Travel Angle. The arc-tangent value between a vehicle’s current and recent past location is measured as “Travel Angle” of the vehicle at a particular moment. This calculated Travel Angle of vehicles is compared to determine whether they are on same road or different. Vehicles traveling on same road are given more priority as next hop node than others. But this algorithm of relevance calculation has few major limitations which are enlisted below:

- It can not distinguish whether two vehicles are traveling along same road or they are traveling along two parallel roads.
- It does not take into account whether a vehicle is moving towards the incident location or moving away from the location.

These issues have been addressed in the enhanced algorithm VRDD which is discussed in next section. A feature wise comparison among exiting algorithms and VRDD are shown in Table 1.

## 4 Proposed VRDD Algorithm

When an emergency incident happens within AOI of VANET, some vehicles, depending on their travel path and the incident location, need to respond quickly to the incident whereas some vehicles can ignore the incident. An example of this scenario has been described in Section 1.1. In VRDD, we present a heuristic method to predict the affectedness or *relevance* of a vehicle and calculate desynchronized rebroadcast delay to control flooding. The detailed methodology of VRDD involving various steps viz. (1) Travel Path Line Estimation, (2) Vehicle Relevance Prediction, (3) Suitability Index and Re-Broadcast Delay Calculation, (4) Network Bridge Vehicles Selection, and (5) Data Dissemination Challenges Handling are described in the following subsections.

### 4.1 Travel Path Line estimation:

In the proposed VRDD, each vehicle at any moment keeps track of its last five GPS positions, which is maintained in a queue of size five. The position sample frequency is proportional to its current

**Algorithm 1** “Least Square” method used for calculating Slope  $M$  and Y-Intercept  $C$  of Travel Path Line

**Input:** Stored last five positions of vehicle

**Output:** Slope  $M$  and Y Intercept  $C$

```

// Calculate the mean of  $y$  as  $\bar{Y}$ :
1:  $\bar{Y} = \frac{1}{n} \sum_{i=1}^5 y_i$ 
// Calculate the mean of  $x$  as  $\bar{X}$ :
2:  $\bar{X} = \frac{1}{n} \sum_{i=1}^5 x_i$ 
// Get the Slope  $M$  :
3:  $xSqrError = \sum_{i=1}^5 (x_i - \bar{X})^2$ 
// Safeguard against infinite or extreme large slope by setting upper
  cap of slope 500
4: if  $xSqrError < 0.1$  then
5:    $xSqrError = 0.1$ 
6: end if
7:  $M = \frac{\sum_{i=1}^5 (x_i - \bar{X})(y_i - \bar{Y})}{xSqrError}$ 
8: if  $M > 500$  then
9:    $M = 500$ 
10: end if
// Get the Y-Intercept  $C$ 
11:  $C = \bar{Y} - M \cdot \bar{X}$ 

```

velocity such that a vehicle is able to generate five samples within a distance of approx 200 meters which means, one position sample is taken in every 40 meters of travel distance. When a vehicle receives a SM, it applies the well known “Least Squares” method on its stored last five positions to get the best fit line which represents the **Travel Path Line (TPL)** of that vehicle at the time of receiving the SM. By this method, path estimation error due to steering movement along a road for lane change or to avoid obstacle can be minimised. Slope  $M$  and Y-Intercept  $C$  of  $TPL$  is calculated as shown in Algorithm 1. By empirical analysis with several topology, we have found that the direction of a road usually doesn’t change much in a span of 40 meters. So, more sample points will unnecessarily result increased computation overhead in the least square method to estimate TPL. On the other hand, if we try to accumulate more samples for TPL estimation by increasing the span of sampling distance greater than 200 meter, it will be too much of generalisation for TPL estimation so that TPL of two different roads will tend to be similar.

### 4.2 Predicting Relevance:

We define the **Relevance** ( $R$ ) of a vehicle by its likelihood of reaching nearer to its immediate previous hop vehicle of SM. The value of  $R$  ranges from 0 to 1, where 1 denotes the highest relevance. The relevance  $R$  depends on the acute angle  $\theta$  between the  $TPL$ s of its immediate previous hop sender vehicle and itself where  $\theta$  is calculated by following equation.

$$\theta = \tan^{-1} \left( \frac{M_s - M_r}{1 + M_s \cdot M_r} \right) \quad (1)$$

$M_s$  and  $M_r$  of Eq. 1 are the tangent ( $M$ ) of sender’s  $TPL$  and receiver’s  $TPL$  respectively and is calculated by Algorithm 1. After

**Algorithm 2** Algorithm for calculating Relevance  $R$  in VRDD

**Input:** Sender's Slope  $M_s$ , Sender's Y-Intercept  $C_s$ , Sender's Position  $P_s(x_s, y_s)$ , Receiver's Slope  $M_r$ , Receiver's Y-Intercept  $C_r$ , Receiver's current Position  $P_r(x_r, y_r)$ , Receiver's fifth last Position  $P_{r5}(x_{r5}, y_{r5})$

**Output:** Relevance  $R$  of Receiver

```

1: Initialize  $\theta_t, C_t$ 
   // Calculate the acute angle  $\theta$  between the Travel Path Line of sender
   and receiver:
2:  $\theta = \tan^{-1}(\frac{M_s - M_r}{1 + M_s \cdot M_r})$ 
3: if  $\theta > 90$  then
4:    $\theta = 180 - \theta$ 
5: end if
   // Check if sender vehicle and receiver vehicle are on same road:
6:  $\Delta C = \text{abs}(C_s - C_r)$ 
7: if  $\theta < \theta_t$  AND  $\Delta C < C_t$  then
8:    $R = 1$ 
9: else
10:   $D_0 = \sqrt{(x_s - x_r)^2 + (y_s - y_r)^2}$  //Current distance from sender
11:   $D_5 = \sqrt{(x_s - x_{r5})^2 + (y_s - y_{r5})^2}$  //Fifth last distance from
   sender
   // Check whether moving towards or away from sender
12:  if  $D_0 < D_5$  then
13:     $R = 0.5 + 0.5 \frac{\theta}{90}$ 
14:  else
15:     $R = 0.5 \frac{\theta}{90}$ 
16:  end if
17: end if

```

receiving a SM when a receiver vehicle calculates  $\theta$ , it sets the value of  $M_r$  with its own calculated  $M$  and obtains  $M_s$  readily from the SM which is provided by sender. The relevance also depends on the Y-Intercept difference  $\Delta C$ . As shown in Algorithm 1, if two neighbor vehicles are travelling along the same road, they will have similar  $TPLs$  and the acute angle  $\theta$  between them will be below a threshold angle tolerance  $\theta_t$ . In the case when two vehicles are traveling along different roads which are nearly parallel to each other,  $\theta$  produced between them will be below  $\theta_t$  but the difference between Y-Intercepts or  $\Delta C$  will be higher than a threshold  $C_t$ . The detailed procedure to calculate  $R$  is given in the Algorithm 2. As per the algorithm, the receiver vehicles traveling along with the sender vehicle's road will have  $R \approx 1$ . If they are on different roads and their travel paths are near to be perpendicular to each other, we assume that the roads on which the vehicles are traveling are going to intersect in a nearby location. In that case, if the vehicle is moving towards the sender vehicle's location, the algorithm assigns higher  $R$  value to it than a vehicle moving away from the sender vehicle. Vehicles traveling along roads which are almost parallel to the sender's road, are assigned lowest  $R$  by the algorithm.

The above mentioned method has the limitation in case of sharp bend or U turn. In this case, vehicles on same road will have very different  $TPL$  from each other and will assume themselves to be on different roads. But we are becoming optimistic here by an realistic assumption that there will be very less number or no sharp bending and U turn inside the  $AOI$  of  $SM$  generation place and even if one is observed, only a very few vehicle at that moment will cross those sharp bends. As only a few vehicle will make their calculations wrong and as every vehicle's calculation is independent of each other, these few wrong calculations will have minimal effect on this algorithm.

To disseminate a SM towards relevant vehicles the vehicles with higher  $R$  needs to be selected as next hop rebroadcast nodes. But it may happen that the vehicles who are closer to sender, are always selected as next hop vehicles. This makes the dissemination slow, as in each hop SM is traversing only a small distance. For faster dissemination, furthest vehicle needs to be selected as rebroadcasting node in each hop. To balance the relevance and fastness trade-off, we have defined a new parameter named "*Suitability Index*" combining relevance factor and fastness factor. VRDD algorithm uses this Suitability Index to calculate Re-Broadcast delay for each vehicle, by which they contest among themselves to get chance to rebroadcast the SM. The procedures of calculating Suitability Index and Re-Broadcast Delay are described in the following Subsection.

### 4.3 Suitability Index and Re-Broadcast Delay Calculation

"*Suitability Index*" ( $S_i$ ) of a vehicle ranges from 0 to 1, denotes how much suitable a vehicle is to be the next rebroadcast node of a SM. Each vehicle calculates its own  $S_i$  independently as given below:

$$S_i = \alpha \cdot R + \beta \cdot D_{norm} \quad (2)$$

$D_{norm}$  of Eq. 2 is the normalized distance with respect to the communication radius ( $R_c$ ) between sender and receiver vehicles, which ranges in [0-1]. We balance between relevance and fastness of dissemination by using two tuning variables  $\alpha$  and  $\beta$  such that  $\alpha + \beta = 1$ . If we want to give priority to relevance of dissemination, we need to set higher value of  $\alpha$ . But that can make dissemination process slow because nearby vehicles from sender may be selected in each hop for dissemination. To increase the fastness of dissemination of message, we need to set higher value of  $\beta$  which may cost losing the relevance of dissemination. The  $S_i$  is used to calculate the "*Re-Broadcast Delay*" ( $T$ ) for a vehicle and is calculated in the following steps:

(i) **Normalized Neighbor Count calculation:** Neighbor count ( $N$ ) of a vehicle is normalized with respect to a value  $N_{max}$  (Maximum Neighbor count) to get "Normalized Neighbor Count" ( $N_{norm}$ ) which ranges from 0 to 1 and used to measure the network density surrounding a vehicle. With the help of received periodic beacon message, a vehicle keeps track of its current neighbor count. The value of  $N_{max}$  is proportional with the transmission power of the vehicles. Greater value of transmission power causes a vehicle receiving beacons from more vehicles and hence increases  $N_{max}$ . The value of  $N_{norm}$  is calculated as:

$$N_{norm} = \frac{N}{N_{max}} \quad (3)$$

(ii) **Density Factor calculation:** Density Factor ( $T_f$ ) is used to adjust the rebroadcast delay of a vehicle depending on the current density of topology surrounding that vehicle. The vehicles with higher  $S_i$  calculates shorter rebroadcast delays for themselves in comparison to the vehicles with lower  $S_i$ . But, if the topology becomes too dense, many nearby vehicles will have almost same  $S_i$  and hence will calculate same rebroadcast delay for themselves which will result in packet collision. So to de-synchronize their rebroadcast delay, we apply Density Factor while calculating "Rebroadcast Delay". The equation for calculating  $T_f$  is given below:

$$T_f = (T_H)^{N_{norm}} \quad (4)$$

The term  $T_H$  in Eq. 4 is Delay Tolerance per hop which denotes the max time (in ms) a vehicle is allowed to wait before rebroadcasting a SM. The value of the  $T_H$  can be set depending how critical the message is. The value of  $T_f$  of Equation 4 lie between 1 ms to  $T_H$  ms and increases exponentially with  $N_{norm}$  which helps to desynchronize rebroadcast schedule with increasing vehicle density.

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**Algorithm 3** Algorithm for calculating Suitability Index  $S_i$  and Rebroadcast Delay  $T$ 


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**Input:** Relevance  $R$  of a vehicle, Current Distance  $D_0$  from sender, Number of Neighbors  $N$ , Communication Radius  $R_c$ , Max Neighbor Count  $N_{max}$ , Delay Tolerance per hop  $T_H$

**Output:** Rebroadcast Delay  $T$  in ms

---

```

// Calculate the normal distance ( $D_{norm}$ ) from the sender:
1:  $D_{norm} = \frac{D_0}{R_c}$ 
2: Initialize  $\alpha = 0.7$  and  $\beta = 0.3$ 
// Calculate the Suitability Index ( $S_i$ ):
3:  $S_i = \alpha.R + \beta.D_{norm}$ 
// Calculate normalized Neighbor Count  $N_{norm}$ :
4:  $N_{norm} = \frac{N}{N_{max}}$ 
// Calculate Density Factor  $T_f$ :
5:  $T_f = (T_H)^{N_{norm}}$ 
// Calculate Rebroadcast Delay  $T$ :
6:  $T = (1 - S_i).T_f$ 

```

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412 (iii) **Rebroadcast Delay calculation:** “Rebroadcast Delay”  $T$  is the  
413 time a vehicle waits before it rebroadcasts a received SM.  $T$  is  
414 calculated as below:

$$T = (1 - S_i).T_f \text{ ms} \quad (5)$$

415 Eq. 5 shows that for vehicles with high  $S_i$ , the re-transmission  
416 delay is very low. The vehicle with very low  $S_i$  waits approx  
417  $T_f$  ms time before it can rebroadcast a SM. If the topology is  
418 very sparse,  $T_f \approx 1$  which ensures fast dissemination even by  
419 vehicle of low suitability index.

420 The detailed method of calculating  $S_i$  and  $T$  is provided in  
421 Algorithm 3.

#### 4.4 Network Bridge Vehicles selection:

423 When the topology is fragmented among small disconnected net-  
424 works of vehicles, one or more vehicle(s) needs to perform the role  
425 of “Network Bridge (NB)” to disseminate the SM from one network  
426 fragment to another. For this, VRDD aims to select the farthest vehi-  
427 cle along every road as NB. After receiving a SM, a vehicle decides  
428 by itself whether it has to play the role of NB in two steps. First, it  
429 checks its neighbor list and selects the neighbors whose  $TPL$  is sim-  
430 ilar to itself (traveling on same road). In next step, it checks whether  
431 its own distance from the SM sender vehicle is the farthest among the  
432 selected neighbors’ distance from SM sender vehicle. If that is true, it  
433 will act as NB. Upon receiving a Beacon Message(BM) from a new  
434 vehicle, it checks the position,  $TPL$  Slope and  $TPL$  Y-Intercept  
435 included in BM of the new neighbor. Based on those parameters it  
436 determines whether the new vehicle is traveling along the same road  
437 and is further away from the SM sender vehicle than its own distance  
438 from SM sender vehicle. If yes, it immediately rebroadcasts the SM  
439 and goes to idle state. The algorithm for selecting NB vehicles is  
440 given in Algorithm. 4.

#### 4.5 Handling Dissemination Challenges in VANET:

442 In VRDD, each vehicle broadcasts periodically “Beacon Message”  
443 (BM) containing its current position,  $TPL$  Slope and  $TPL$  Y-  
444 Intercept. All vehicles are aware of their neighborhood by main-  
445 taining a neighbor list. If a vehicle experiences or observes any  
446 emergency incident, it originates and broadcasts a *Safety Message*  
447 (SM) which contains the information about the incident, a unique  
448 message id, incident location, its own current position,  $TPL$  Slope  
449 and  $TPL$  Y-Intercept. On receiving a SM, each receiver vehicle  
450 predicts its own Relevance (refer Subsection 4.2) and schedules a

---

**Algorithm 4** Algorithm for selecting Network Bridge vehicle

---

**Input:** Neighbor SET  $N$ , Sender’s Position  $P_s(x_s, y_s)$  and Receiver’s  
current Position  $P_r(x_r, y_r)$

**Output:** Boolean flag NETWORK\_BRIDGE (TRUE if selected as Net-  
work Bridge, else FALSE)

---

```

// Get the subset of  $N$  which are on same road of receiver vehicle and
store in set  $\{N_{sr}\}$ 
1: Initialize  $\theta_t = 5$ ,  $C_t = 50$  and NETWORK_BRIDGE =
TRUE
// Calculate the acute angle  $\theta$  between the Travel Path Line of the
receiver vehicle and its neighbors:  $M_r$  is receiver vehicle path slope
and  $M_{Nbr_i}$  is  $i^{th}$  neighbor’s path slope
2: for all  $Nbr_i$  in  $N$  do
3:  $\theta = \tan^{-1}(\frac{M_r - M_{Nbr_i}}{1 + M_r.M_{Nbr_i}})$ 
4: if  $\theta > 90$  then
5:  $\theta = 180 - \theta$ 
6: end if
// If receiver vehicle and  $Nbr_i$  are on same road, add  $Nbr_i$  into
the set  $\{N_{sr}\}$ .  $C_r$  is receiver’s path y-intercept and  $C_{Nbr_i}$  is  $i^{th}$ 
neighbor’s path y-intercept
7: if  $\theta < \theta_t$  AND  $abs(C_r - C_{Nbr_i}) < C_t$  then
8:  $Nbr_i \rightarrow \{N_{sr}\}$ 
9: end if
10: end for
// Calculate the distance of current vehicle from sender:
11:  $D_r = \sqrt{(x_s - x_r)^2 + (y_s - y_r)^2}$ 
// Check whether it is the last vehicle among the neighbors
12: for all  $Nbr_i$  in  $\{N_{sr}\}$  do
13:  $D_{Nbr_i} = \sqrt{(x_s - x_{Nbr_i})^2 + (y_s - y_{Nbr_i})^2}$ 
// If any neighbor  $N \in \{N_{sr}\}$  is at farther distant from sender than
receiving vehicle the receiver will not act as Network Bridge.
14: if  $D_{Nbr_i} > D_r$  then
15: NETWORK_BRIDGE = FALSE
16: break
17: end if
18: end for

```

---

rebroadcast with the timer of calculated delay (refer Subsection 4.3).  
Before the timer fires, if a vehicle hears a SM with the same inci-  
dent id, it discards its schedule assuming that the SM has been  
disseminated unless it has decided to act as Network Bridge (NB).  
Thus “Broadcast Storm” problem is mitigated in VRDD by selec-  
tive rebroadcast by relevant vehicles and also the SM is disseminated  
along the relevant directions with higher priority. In case a vehicle  
decides to act as NB as per the method mentioned in Subsection 4.4,  
it waits to meet a suitable vehicle from any other network fragment to  
disseminate the SM. Thus, “Network Partition” problem is handled.  
Flowchart of VRDD algorithm is described in Fig. 3.

#### 4.6 Complexity Analysis:

VRDD is a distributed algorithm which will run on all vehicles  
inside a topology. The input size of VRDD algorithm therefore  
will be the number of neighbors (N) of each vehicle. A vehicle  
calculates its relevance by Algorithm 1 and Algorithm 2. Then it cal-  
culates its rebroadcast schedule by Algorithm 3. It can be seen that  
Algorithm 1, Algorithm 2 and Algorithm 3 doesn’t depend on the  
neighbor count of vehicles and performs a fixed amount of compu-  
tations. Hence their time complexity is  $O(1)$ . Whereas a vehicle runs  
Algorithm 4 to handle network partition issue which is dependent on  
neighbor count. There are two loops in line 2 and 12 in Algorithm

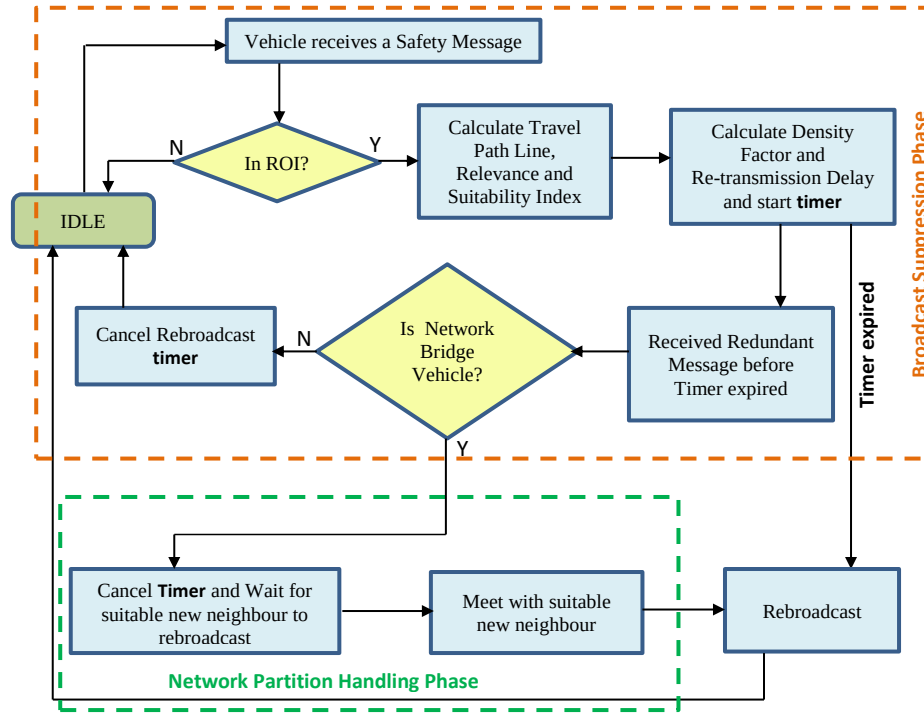


Fig. 3: Flow Chart of VRDD algorithm

473 4 both of  $O(N)$ . Hence, the overall time complexity of network partition handling is  $O(N)$ . It is noted that while fastness is one of the primary evaluation metric of data dissemination, network partition handling algorithm is evaluated only by its effectiveness i.e, the ratio of vehicles inside AOI which receives SM.

## 478 5 Performance Evaluation

479 This section describes the simulation setup and analyze the performance of the proposed VRDD algorithm in comparison to some existing benchmark methods on various performance metrics.

### 482 5.1 Simulation Topology

483 We have considered two different types of topology representing urban scenarios, and one type of topology representing highway scenario for evaluating the performance of our proposed VRDD algorithm. The highway scenario topology has almost straight road with no major turn or junction. We have used three such topology in our simulation and one of them is given in Fig. 4 which is taken from I 70; US 24, Arriba, Lincoln County, United States of America. First type of urban scenario is of irregular shape where no geometric pattern can be observed. We have used three topology of this type for our simulation and one of them is shown in Fig. 5 which is taken from Kings Park West, Fairfax Country, Virginia. Second type of urban topology is of grid type. We have considered three topology of this type for our simulation and one of them is shown in Fig. 6 which is taken from 1st A Cross road, GC block, Bidhannagar, Kolkata 700102, West Bengal, India. In all scenarios, junctions are regulated by automatic traffic signals. In each of the scenarios, we have varied the densities of vehicles into four categories namely sparse, normal, dense and highly dense. All the vehicles originates in 508 the topology with zero speed, then accelerate with 2.4 m/s till max speed of 12 m/s (43km/hr approx). But in its way it may halt due 509 to traffic signal in junction points. Vehicles on lower priority roads 510 have to wait in the junction to allow vehicles on higher priority roads 511 to pass by. The junction points of the topology are auto traffic reg- 512 ulated. For the highway scenarios we considered that there are no 513 junction points. 514

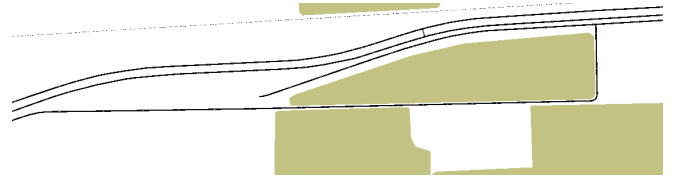


Fig. 4: Highway topology from I 70; US 24, Arriba, Lincoln County, Colorado, USA

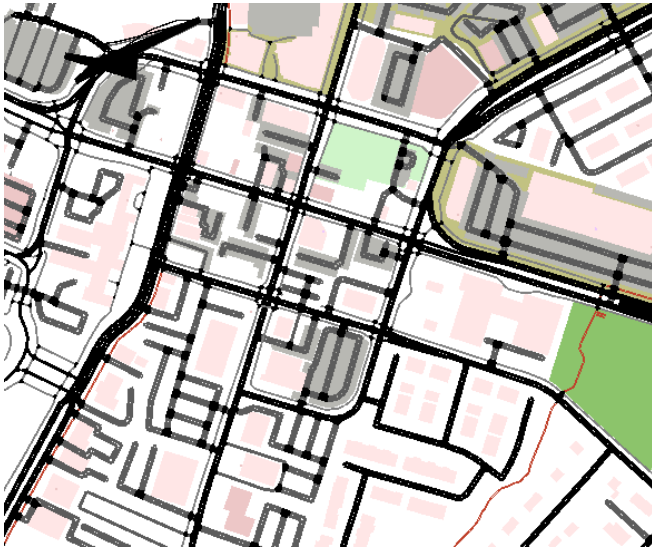
Table 2 Simulation Parameters

| Parameters                     | Values                 |
|--------------------------------|------------------------|
| AOI Area                       | 1 km <sup>2</sup>      |
| Transmission Power of Vehicles | 5mW                    |
| Transmission Bit Rate          | 6 Mbps                 |
| Beacon Message Interval        | 1 sec                  |
| Simulation Time                | 800 sec                |
| Beacon Message Carrier         | Channel 172 (5.86 GHz) |
| Safety Message Carrier         | Channel 172 (5.86 GHz) |
| $\alpha, \beta$                | 0.7, 0.3               |
| $\theta_t, C_t$                | 5, 50                  |
| $N_{max}, T_H$                 | 100, 20                |

### 5.2 Simulation Environment

The network simulator OMNET++ 5.1 [20] and UBUNTU 16.04 operating system are used to run the simulations. We have used SUMO 0.30.0 [21] to run the vehicle traffic flows and Veins 4.6 [22] framework is used for connection between OMNET++ and SUMO. The values of simulation parameters used in our experimentation are given in Table 2:





**Fig. 5:** Irregular urban topology from Kings Park West, Fairfax Country, Virginia



**Fig. 6:** Grid urban topology from 1st A Cross road, GC block, Bidhannagar, Kolkata

### 5.3 Performance Metrics

We have evaluated the performance of the proposed VRDD Algorithm in comparison to other benchmark algorithms on various performance metrics (viz. Packet Loss (PL), Packet Redundancy (PR), Dissemination Delay (DD), and Vehicle Coverage Ratio (VCR)) as used in other existing benchmark methods. In this paper, we have also introduced a new performance metric namely Relevant Vehicle Penetration Ratio (RVPR) to determine the effective reception of Safety Message to the targeted relevant vehicles.

(i) **Packet Loss (PL):** It is the total packet loss on service channel (SCH channel number 172) for disseminating a Safety Message within an AOI of the topology.

(ii) **Packet Redundancy (PR):** This is measured by the average duplicate Safety Message received by any vehicle. This indicates how effectively and efficiently next hop node is chosen for rebroadcasting the SM. Higher value of PR denotes that the SM is not propagating quickly or the broadcast storm suppression

mechanism is not behaving properly. It is computed as:

$$PR = \frac{\text{Total number of duplicate SM received by all vehicles}}{\text{Total number of informed vehicles}}$$

(iii) **Dissemination Delay (DD):** This represents the average delay of receiving the Safety Message for the first time by any vehicle given by milli seconds. It is computed as:

$$DD = \frac{\text{Total delay of receiving SM first time by all vehicles}}{\text{Total number of informed vehicles}}$$

(iv) **Vehicle Coverage Ratio (VCR):** This metric denotes the ratio (given in percentage) between number of informed vehicles and total number of vehicles present in scenario at the time when Safety Message originated. This metric is useful to evaluate how effectively the “Network Partition” problem is handled by an algorithm. It is computed as:

$$VCR = \frac{\text{Total number of informed vehicles}}{\text{Total number of vehicles}}$$

(v) **Relevant Vehicle Penetration Ratio (RVPR):** We have introduced this metric to evaluate the overall effectiveness of a method for Safety Message dissemination to the targeted relevant vehicles. RVPR is computed as:

$$RVPR = \frac{\text{Total number of relevant vehicles informed}}{\text{Total number of targeted relevant vehicles}}$$

As in Highway Scenario, all vehicles are relevant, this metric is not applicable in that case. This metric is applicable only to urban scenarios. For computation of this metric in simulation, we keep the targeted relevant vehicles in a list for a particular safety message.

### 5.4 System Assumptions

Some of the assumptions considered for the system during simulation are as follows:

- We have considered that there is no channel error. Packet collision is the only reason for packet loss.
- In safety channel 172 (5.86 GHz), only two types of packets are transmitted namely periodic safety beacon message (BM) and emergency safety message (SM).
- We have also considered the fact that as dissemination delay is bounded by few milliseconds, the duration of dissemination falls between two successive periodic beacon message and thus beacon message has no role in collision. Otherwise it would be difficult to estimate whether a packet loss is due to inefficient scheduling of the protocols or due to channel error or beacon message collision.
- Only one accident happens in the simulation duration and hence only one SM will be disseminated among vehicles in the AOI.

In the following Subsection, we have analyzed the simulation results achieved on the above mentioned metrics.

### 5.5 Simulation results and analysis

For evaluating the performance, we have compared our proposed algorithm VRDD with one probability based weighted p-persistence algorithm “DV-CAST” in highway scenario, one delay based sender oriented algorithm “TrAD” and one delay based receiver oriented algorithm “DRIVE” in both highway and urban scenarios. We have simulated these three algorithms along with our proposed VRDD with same simulation parameters in all the scenarios. The metric values are measured against varying vehicle densities. For urban scenario, density values are : 160, 240, 320 and 400 vehicles per  $km^2$  and for highway scenario, density values are : 150, 200, 250 and 300 vehicles per  $km^2$  respectively. In each of the topology, we have generated a Safety Message at ten different simulation times when most of the vehicles are present in the scenario and taken the average of



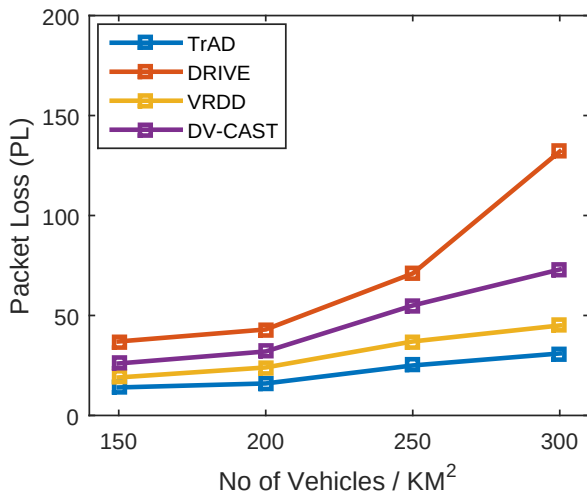


Fig. 7: Packet Loss in Highway Scenario

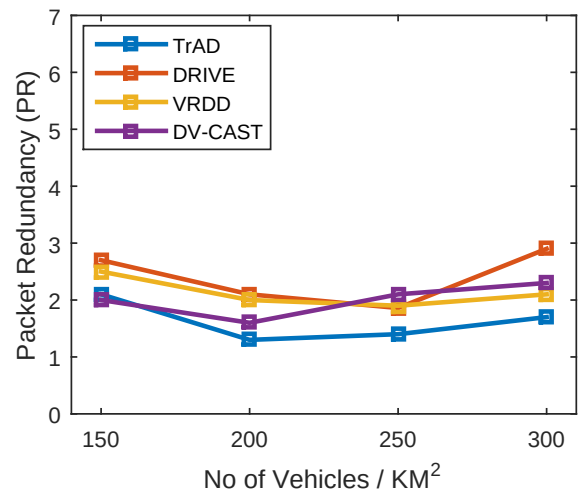


Fig. 9: Packet Redundancy in Highway Scenario

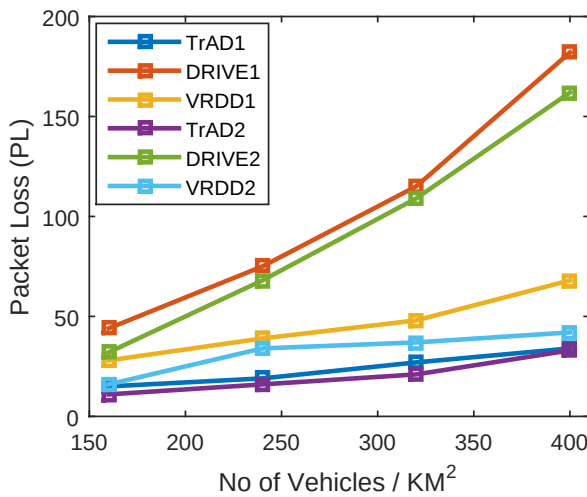


Fig. 8: Packet Loss in Urban Scenario

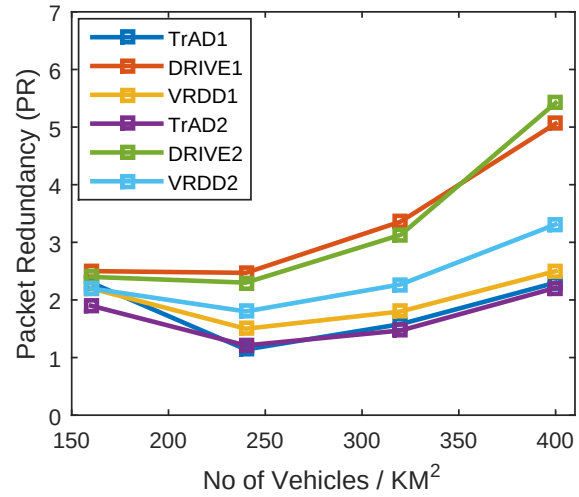


Fig. 10: Packet Redundancy in Urban Scenario

each metrics. Simulation results for highway scenario is given in Fig. 7, 9, 11, 13 and simulation results for urban scenarios are given in Fig. 8, 10, 12, 14, 15. The legends “TrAD1”, “DRIVE1”, “VRDD1” represent the average metric values obtained from three irregular (type 1) topology and “TrAD2”, “DRIVE2”, “VRDD2” represent the average metric values obtained from three grid (type 2) topology. Results analysis on the different metrics are given below:

(i) **Packet Loss (PL):** Fig. 7 shows the comparative results on highway scenario and Fig. 8 shows the comparative results on two urban scenarios for the metric Packet Loss. It can be seen that DRIVE suffers most from the packet loss with the increased vehicle density. In highway scenario for DRIVE, most of the vehicles collectively belong either in sweet spot or in non-sweet spot. Also in dense urban scenario, many vehicles are situated in sweet spot at a given time. So nearby vehicles calculate almost similar dissemination delay for themselves and hence it suffers from packet loss. DV-CAST uses p-persistence method with a fixed waiting time where their probability value is calculated by the normalized distance from sender and because of which, many nearby vehicles can decide to transmit simultaneously which also results to packet loss. VRDD performs better than DRIVE and DV-CAST because it calculates desynchronized rebroadcast delay. Being a sender oriented algorithm, TrAD outperforms all other receiver oriented protocols. The comparative

gain of VRDD over TrAD, DRIVE and DV-CAST are given in TABLE 3, 4 and 5 respectively for the Packet Loss metric.

(ii) **Packet Redundancy (PR):** Fig. 9 shows the comparative results on highway scenario and Fig. 10 shows the comparative results on urban scenario for the metric Packet Redundancy. As it can be seen from the Figures, highway scenario is having less PR value than that of urban scenario because next hop vehicle for retransmission is chosen from only two opposite directions whereas in urban scenario, next hop vehicle for retransmission is chosen from all possible directions and vehicles may hear the same SM from multiple chosen next hops. Also in less dense scenario, algorithms need some extra rebroadcasts to handle network partition issue which increases redundancy ratio. Packet Redundancy metric is also related to Packet Loss metric because when packets get lost, it affects broadcast suppression mechanism which increases redundant rebroadcast. As it can be seen that in dense urban scenario, DRIVE suffers from huge packet loss which results redundant rebroadcasts and hence increases PR metric. VRDD performs better compared to DRIVE and DV-CAST, but lags behind TrAD. In highway scenario, DV-CAST and DRIVE performs better compared to their performance on urban scenarios. The comparative gain of VRDD over TrAD, DRIVE and DV-CAST are given in TABLE 3, 4 and 5 respectively for the Packet Redundancy metric.

**Table 3** Improvement gain of VRDD (in %) over TrAD in urban and highway scenario

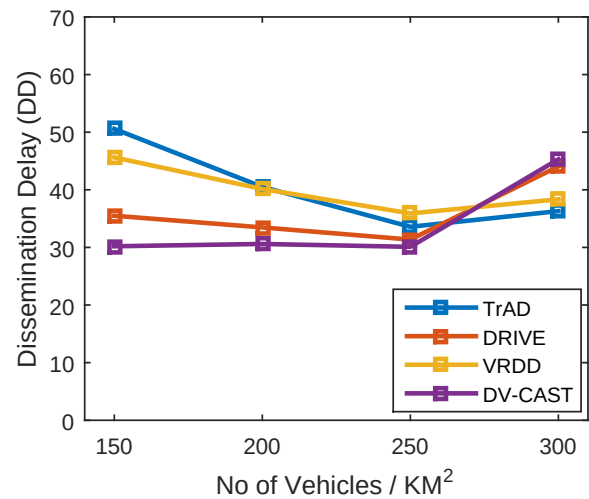
| Metric | No. of Vehicles (highway)/km <sup>2</sup> |        |        |        | No. of Vehicles (urban)/km <sup>2</sup> |        |        |        |
|--------|-------------------------------------------|--------|--------|--------|-----------------------------------------|--------|--------|--------|
|        | 150                                       | 200    | 250    | 300    | 160                                     | 240    | 320    | 400    |
| PL     | -26.31                                    | -33.33 | -32.43 | -31.11 | -46.42                                  | -51.28 | 43.24  | -21.42 |
| PR     | -16                                       | -35    | -26.3  | -19    | 4.5                                     | -24    | -35.24 | -33.23 |
| DD     | 10.8                                      | 0.8    | -6.4   | -5.3   | 4.9                                     | 12     | 4.5    | -6.5   |
| VCR    | 5.5                                       | -0.44  | 2.7    | 1.22   | 2.63                                    | -2.5   | 2.8    | 3.89   |
| RVPR   | NA                                        | NA     | NA     | NA     | 21.4                                    | 16.3   | 25.41  | 16.1   |

**Table 4** Improvement gain of VRDD (in %) over DRIVE in urban and highway scenario

| Metric | No. of Vehicles (highway)/km <sup>2</sup> |       |       |        | No. of Vehicles (urban)/km <sup>2</sup> |       |       |       |
|--------|-------------------------------------------|-------|-------|--------|-----------------------------------------|-------|-------|-------|
|        | 150                                       | 200   | 250   | 300    | 160                                     | 240   | 320   | 400   |
| PL     | 94.73                                     | 79.16 | 91.89 | 193.33 | 57.14                                   | 92.3  | 194.6 | 285.7 |
| PR     | 8                                         | 5     | -2.1  | 38.1   | 13                                      | 64    | 37.88 | 63.74 |
| DD     | -22.2                                     | -16.6 | -12.6 | -14.9  | -8.26                                   | 24.76 | 20.5  | 19.58 |
| VCR    | -3.79                                     | -4.15 | -0.53 | -0.72  | 6.77                                    | 0.70  | 4.17  | 9.63  |
| RVPR   | NA                                        | NA    | NA    | NA     | 12.55                                   | 7.92  | 21.37 | 15.26 |

(iii) **Dissemination Delay (DD):** Fig. 11 shows the comparative results on highway scenario and Fig. 12 shows the comparative results on urban scenario for the metric Dissemination Delay. It can be seen that in sparse scenario, all the algorithms take more time to disseminate data compared to lightly dense scenario. This is because, in sparse scenario, all the algorithms handle “Network Partition” issue where few selected vehicles wait to meet other vehicles from another network fragment to rebroadcast the Safety Message. In lightly dense or moderate dense scenario, vehicles are connected to form a single network and in one transmission, more vehicles get informed about a safety message. Hence, algorithm performs best in these cases. For sparse and lightly dense conditions, in highway scenario, DV-CAST performs best as it has minimum waiting time and computational overhead whereas, in urban scenario, DRIVE performs best because of its low computational overhead. Being a Sender Oriented algorithm, TrAD involves in heavy computation to prepare rank list for all its neighbors which consumes time. VRDD performs better than TrAD but lags behind DV-CAST and DRIVE in sparse and lightly dense scenario. But after that as the vehicle density increases, the dissemination delay strictly increases for all algorithms. In highly dense scenario, both DV-CAST and DRIVE suffer from huge packet loss because of which, some vehicles may get informed via a longer hop route which increases delay. Also, delay increases for TrAD because it gets overburdened with computation to create rank list for large number of neighbors. In dense scenarios, VRDD assigns longer rebroadcast delay to vehicles to avoid packet loss but it also increases delay. It can be seen that in dense urban scenario, VRDD performance is compatible with sender oriented protocol TrAD. The comparative gain of VRDD over TrAD, DRIVE and DV-CAST are given in TABLE 3, 4 and 5 respectively for the Dissemination Delay metric.

(iv) **Vehicle Coverage Ratio (VCR):** Fig. 13 shows the comparative results on highway scenario and Fig. 14 shows the comparative

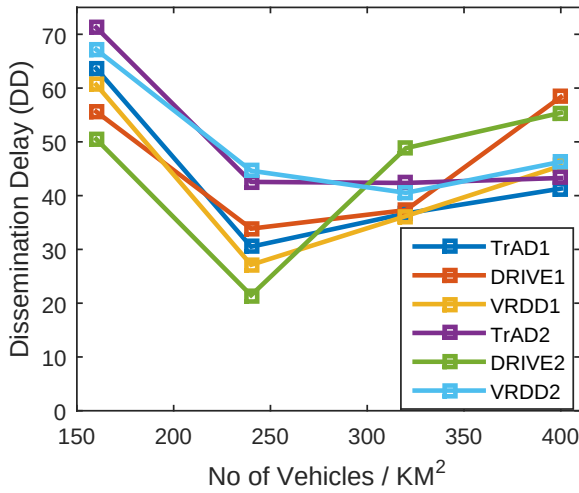
**Fig. 11:** Dissemination Delay in Highway Scenario

results on urban scenario for the metric Vehicle Coverage Ratio. It can be seen that all algorithms overall perform better in highway scenario than in urban scenario in respect to VCR because, in urban scenario, more vehicles need to be covered in many directions whereas, in highway scenario, message needs to travel only one or two opposite directions. Overall we found that the protocols' performances are compatible to each other. The comparative gain of VRDD over TrAD, DRIVE and DV-CAST are given in TABLE 3, 4 and 5 respectively for the Vehicle Coverage Ratio metric.

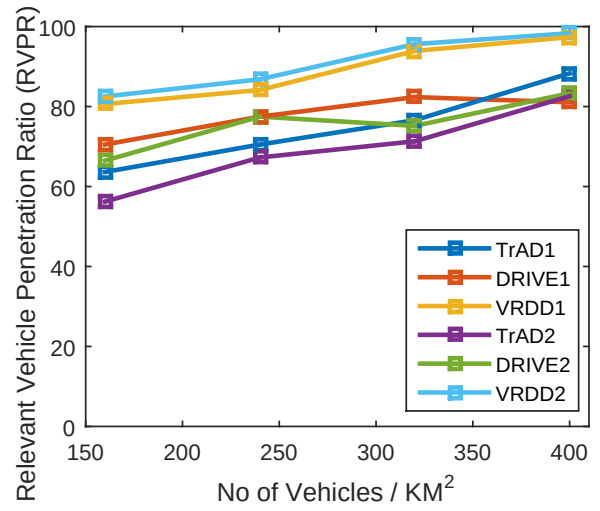
(v) **Relevant Vehicle Penetration Ratio (RVPR):** The calculation of vehicle relevance heavily depends on the value of thresholds  $\theta_t$  and  $C_t$  as mentioned in subsection 4.2. The value of  $\theta_t$  and  $C_t$  is heuristic and after several simulation run on various topology, we observed that in most of the cases, the values of  $\theta$  and  $\Delta C$  between two vehicles of same road are bounded by the values [2-10] and [20-100] respectively. We decided to keep the mode value of the distribution which are:  $\theta_t = 5$  and  $C_t = 50$ . Another important parameter for relevance calculation is the value of  $\alpha$  as mentioned in subsection 4.3. Fig. 16 shows the variation of RVPR metric of VRDD with respect to varying  $\alpha$  with fixed vehicle density 240/km<sup>2</sup> in urban topology. As the  $\alpha$  increases,

**Table 5** Improvement gain of VRDD (in %) over DV-CAST in highway scenario

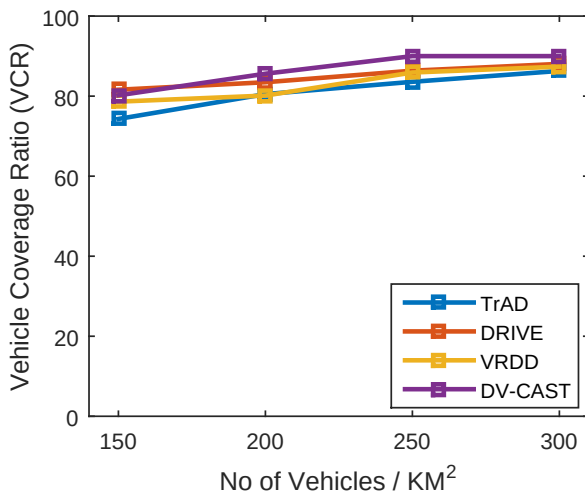
| Metric | No. of Vehicles (urban)/km <sup>2</sup> |       |        |       |
|--------|-----------------------------------------|-------|--------|-------|
|        | 150                                     | 200   | 250    | 300   |
| PL     | 26.31                                   | 33.33 | 48.64  | 62.22 |
| PR     | -20                                     | -20   | 10.53  | 9.52  |
| DD     | -33.8                                   | -23.7 | -16.15 | 18.12 |
| VCR    | -2                                      | -6.81 | -4.77  | -3.81 |



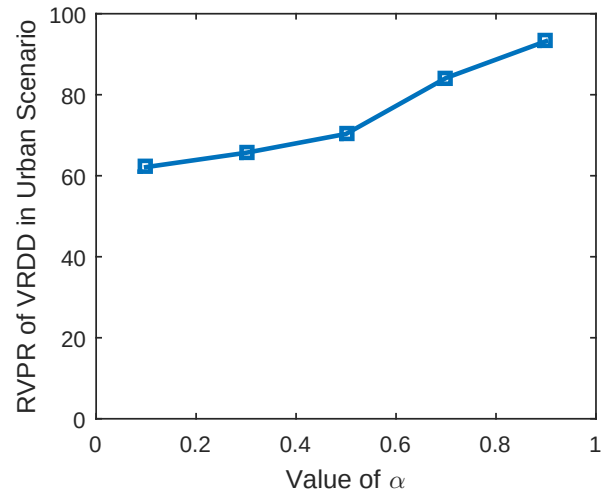
**Fig. 12:** Dissemination Delay in Urban Scenario



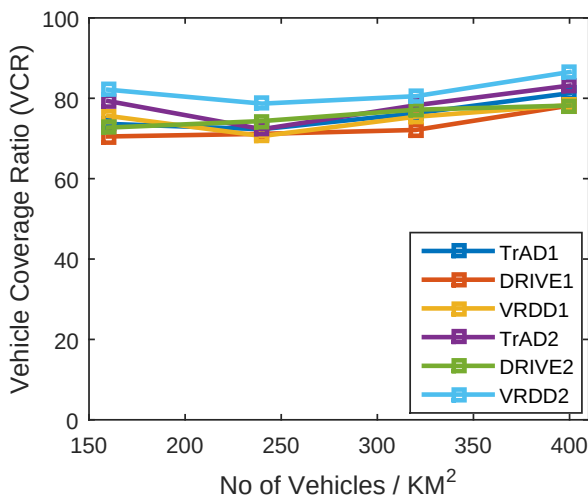
**Fig. 15:** Relevant Vehicle Penetration Ratio in Urban Scenario



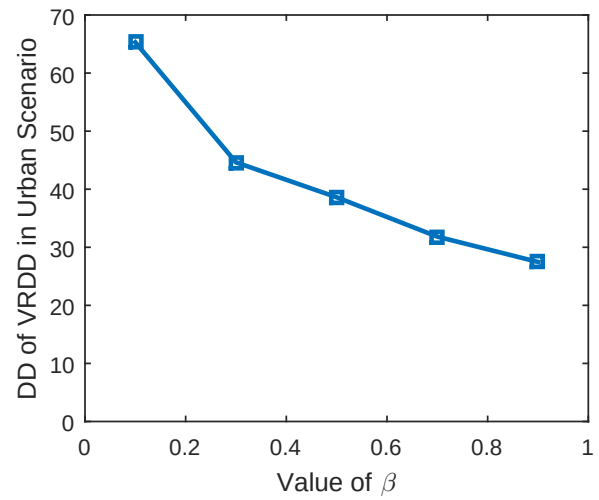
**Fig. 13:** Vehicle Coverage Ratio in Highway Scenario



**Fig. 16:** Variation of metric RVPR wrt  $\alpha$



**Fig. 14:** Vehicle Coverage Ratio in Urban Scenario



**Fig. 17:** Variation of metric RVPR wrt  $\beta$

RVPR also increases. Fig. 17 illustrates the relation between dissemination delay (metric DD) and  $\beta$  with fixed vehicle density

240/km<sup>2</sup> in urban topology. So there is a trade-off between relevance and delay with the value of  $\alpha$  ( $\beta = 1 - \alpha$ ). As we want

to emphasize on relevance property which is the motivation of this work, we choose  $\alpha$  as 0.7 because with this value VRDD can outperform other protocols in RVPR metric. If we increase the value of  $\alpha$  more, the value of  $\beta$  will be less than 0.2 and we observe a sharp rise in the value of delay as shown in Fig. 17. Fig. 15 shows the comparative results on urban scenario for the metric Relevant Vehicle Penetration Ratio. As in highway scenario, all vehicles are relevant, we don't measure this metric in highway scenario. To evaluate this metric in urban scenario, we have a predefined list of vehicles which would be relevant with respect to the Safety Message (SM) origin location and for a given scenario. After simulation, we checked how many of them actually received the SM and we calculated the percentage to get RVPR. Simulation results show that VRDD has the highest RVPR among the compared algorithms because no other algorithms attempt to disseminate data among the directions of relevant vehicles. The comparative gain of VRDD over TrAD, DRIVE and are given in TABLE 3 and 4 respectively for the Relevant Vehicle Penetration Ratio metric.

## 6 Conclusions and Future Work

This paper proposed a novel receiver oriented Vehicular Relevance based Data Dissemination (VRDD) algorithm, aims to handle the two major issues of infrastructure less V2V multi-hop broadcast data dissemination viz. "Broadcast Storm" and "Network Partition", with an additional best effort to disseminate the safety messages towards and by relevant vehicles. Unlike many algorithms, the prerequisite knowledge of road topology (e.g. road ids, junction point list, etc.) or RSU support is not required in VRDD. In this algorithm, each vehicle uses a completely distributive and heuristic method to predict its relevance. Vehicles contest with each other to rebroadcast a safety message where relevant vehicles have greater chance to win the contest. The performance of VRDD is compared with TrAD (sender-oriented scheme), DRIVE (receiver-oriented scheme) and DV-CAST (probability based scheme) on various performance metrics by varying the vehicle density on different topology. Simulation results show that VRDD's performance is compatible with the existing state of art algorithms in terms of packet loss and dissemination delay metrics. However, it is a clear winner on the metric of relevance dissemination. Simulation results also indicate that VRDD is most suitable for dense urban scenarios. However, it has a limitation that it may not select proper relevant vehicles if the road topology includes too much irregular shape like sharp bend or "U" turn.

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